N E X O R A

Azure Enterprise AI Copilot

Conversational analytics / Multimodal RAG / Secure agent orchestration

ENGINEERING PORTFOLIO · PROJECT SHOWCASE

One question. Two evidence sources.

Turning structured commerce data and industry research into one conversational workflow.

THE BUSINESS QUESTION

What do our sales numbers show — and how do they compare with the wider market?

STRUCTURED DATA

Historical Olist e-commerce records in Azure SQL provide calculated category results.

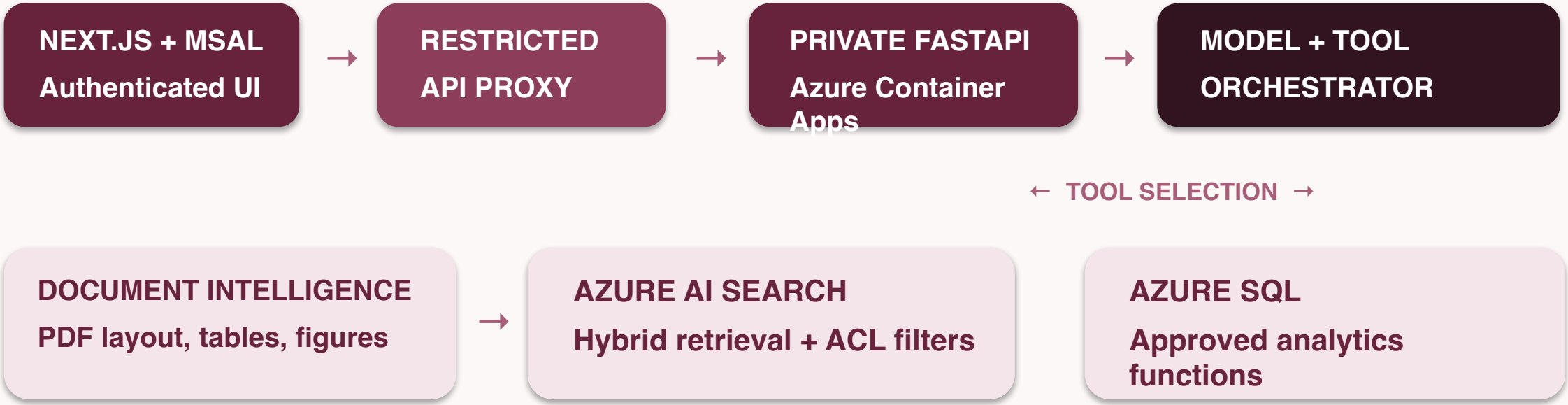
DOCUMENT EVIDENCE

Webshoppers research reports provide qualitative market context and cited pages.

NEXORA connects both, without treating generated text as a source of numerical truth.

Architecture | From question to grounded answer

Actual components represented in the reviewed repository.



Identity validation • Tenant / user / group document filtering • Source references + tool trace

What I built

An integrated system, not a standalone chatbot prototype.

SQL analytics

Approved, parameterized Python tools over historical Olist data.

Multimodal RAG

PDF paragraphs, tables and figures; checkpoint-aware indexing.

Hybrid retrieval

Azure AI Search keyword + vector search with access filtering.

Agent orchestration

Model-guided selection between SQL and document-search tools.

Enterprise identity

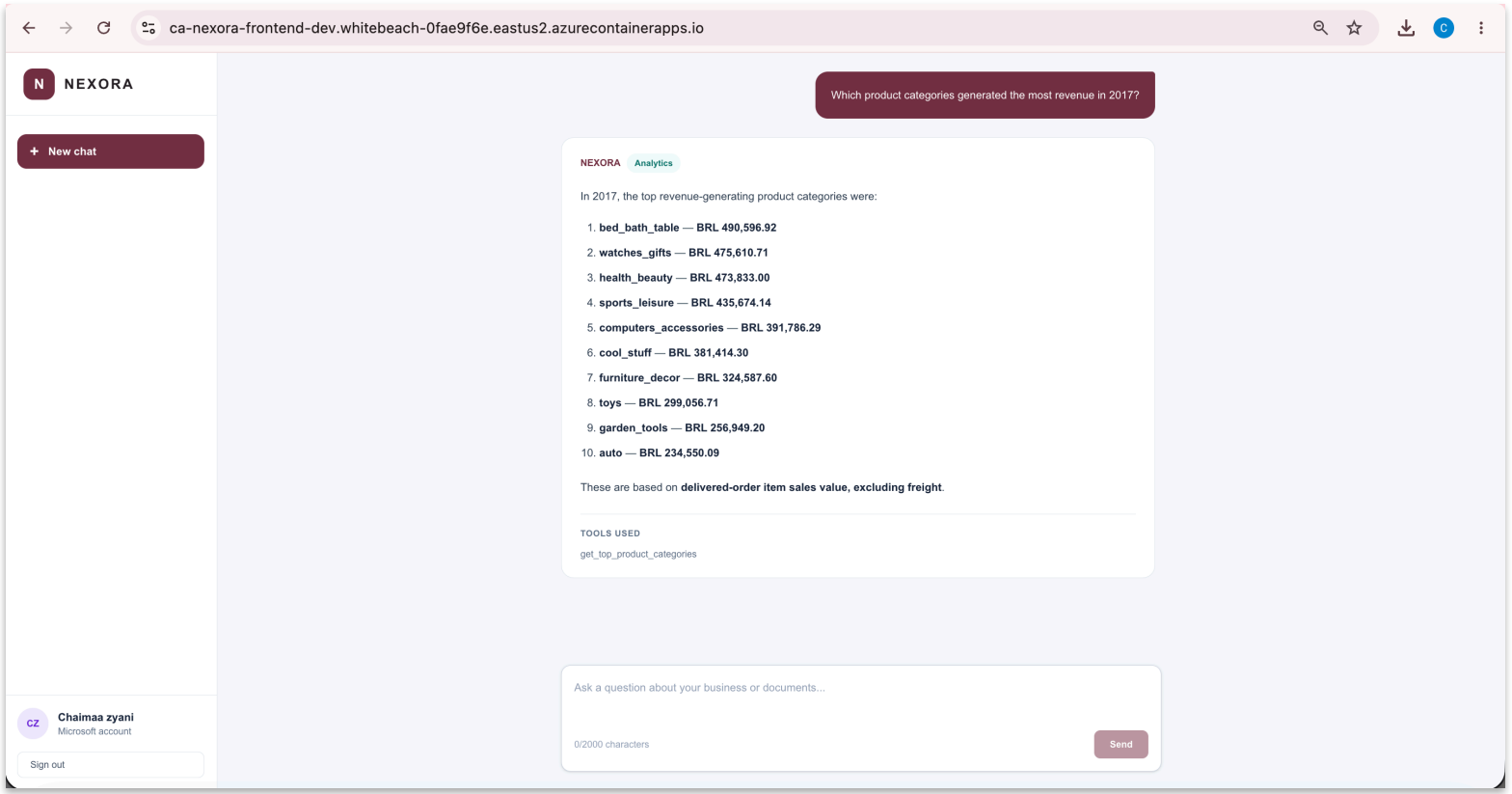
Microsoft Entra token validation, role and scope enforcement.

Cloud delivery

Docker, Container Apps, managed identities, Bicep and GitHub Actions.

Demo 01 | SQL-backed business analytics

The model invokes a registered tool; the numbers come from Azure SQL.



WHAT TO NOTICE

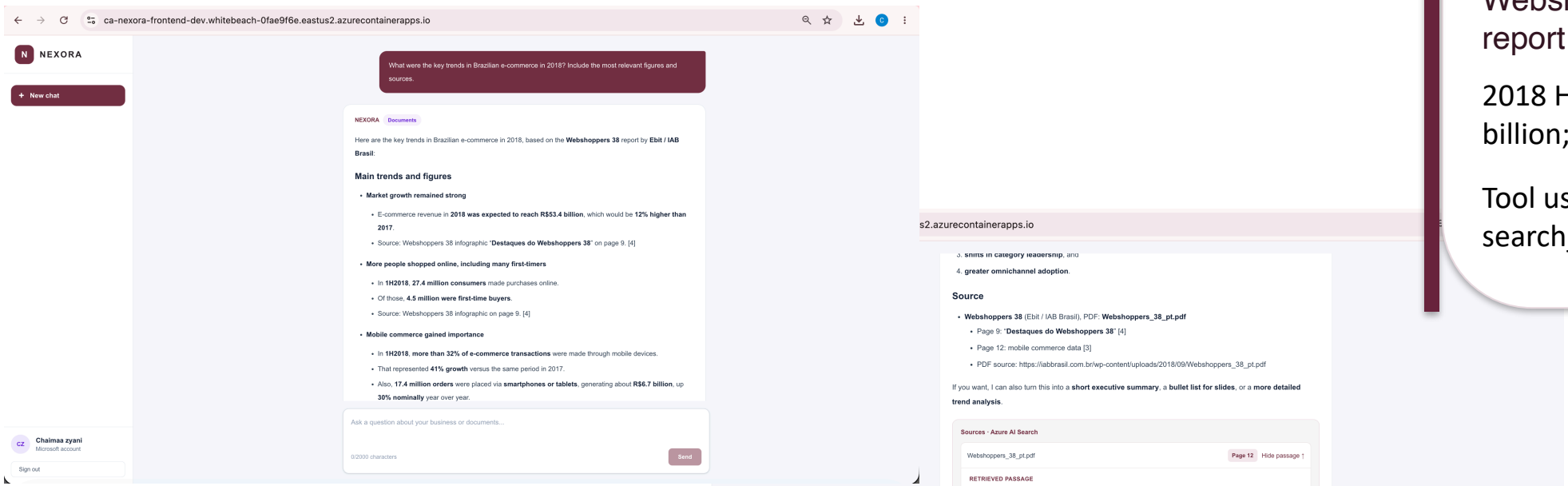
2017 category revenue from Azure SQL.

Top result: bed_bath_table — BRL 490,596.92.

Tool used: get_top_product_categories.

Demo 02 | Document-grounded research

Questions about 2018 market trends use the indexed Webshoppers report.



WHAT TO NOTICE

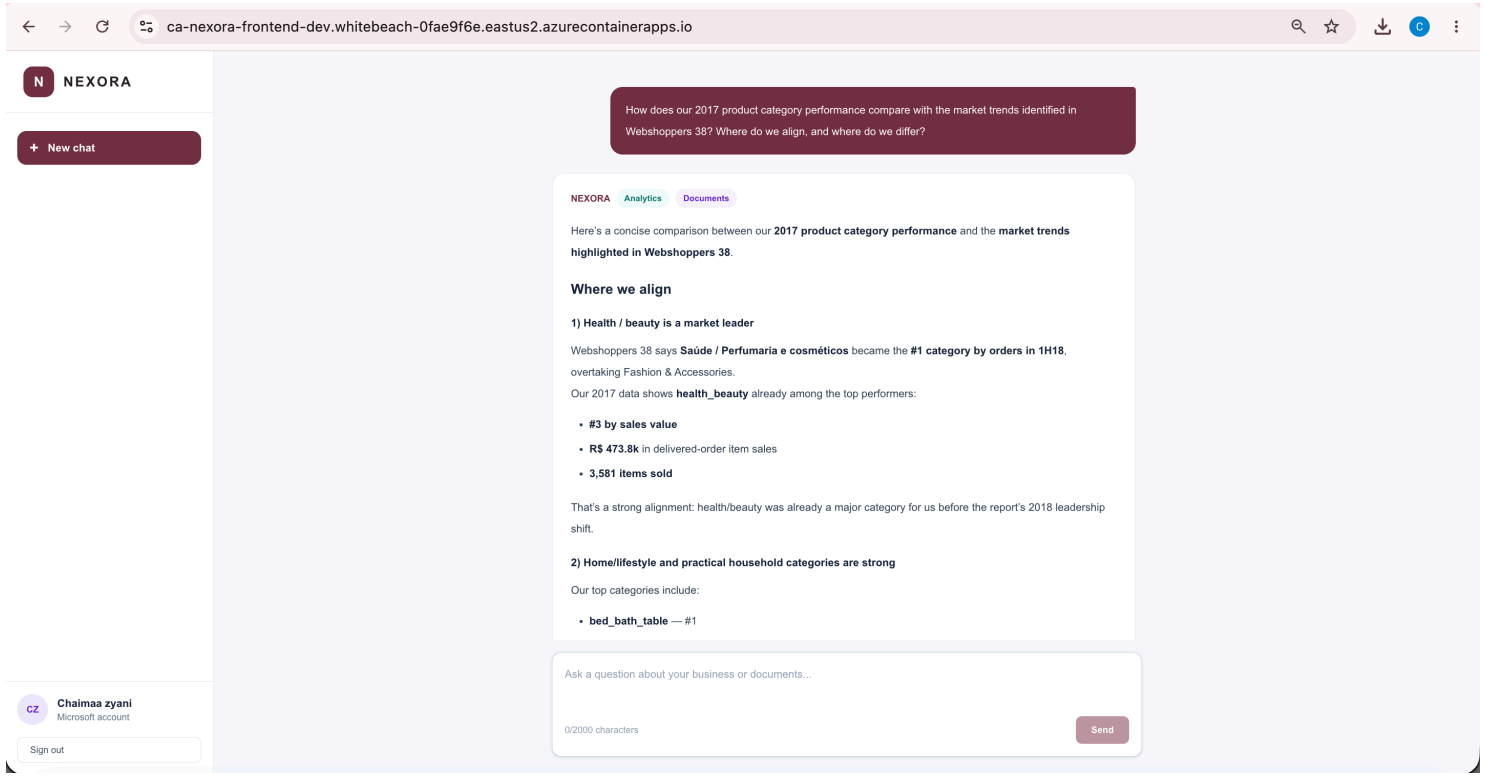
Webshoppers 38 answers with report citations.

2018 H1 e-commerce: R$ 23.6 billion; 27.4 million buyers.

Tool used: search_enterprise_documents.

Demo 03 | Cross-source investigation

One workflow combines SQL category performance with external research.



WHY IT MATTERS

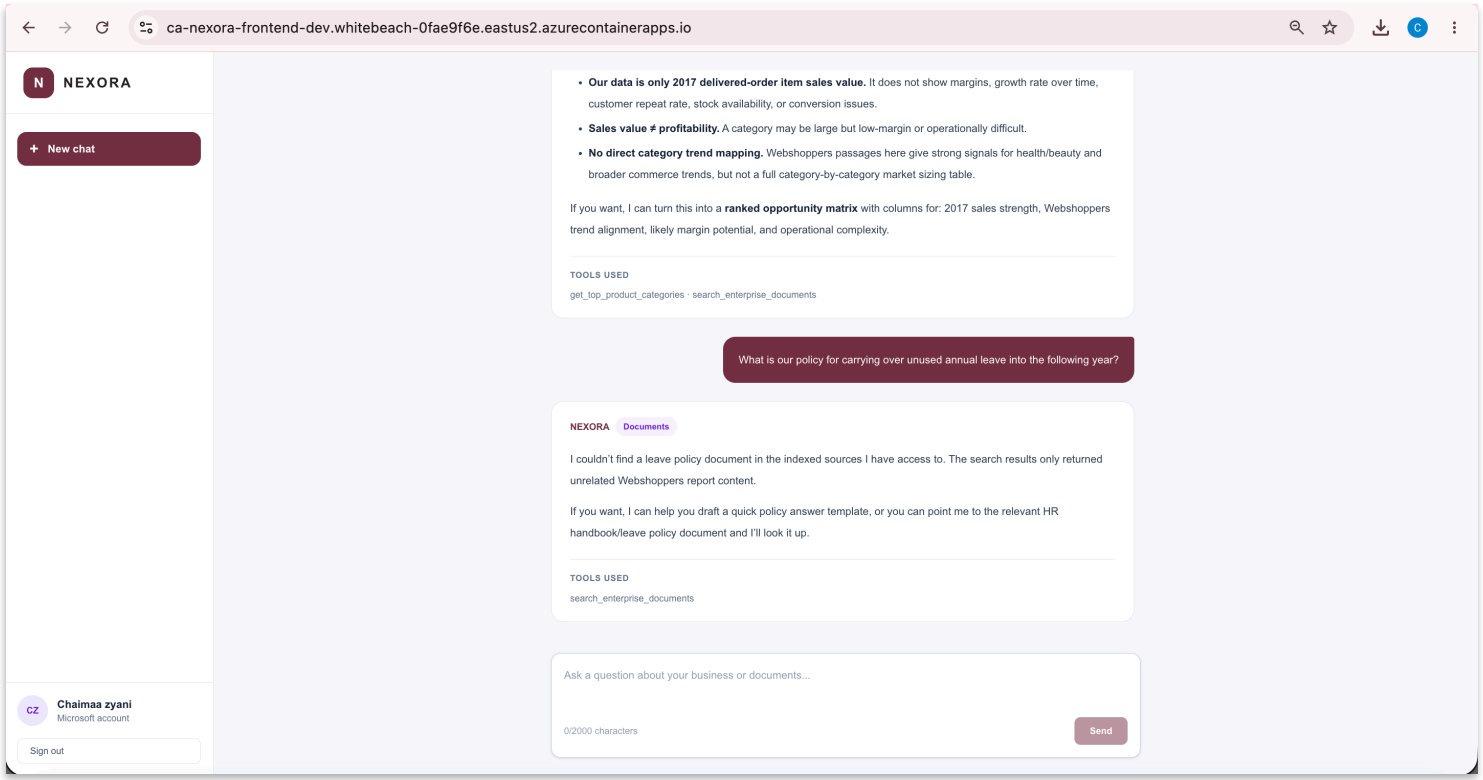
Compares 2017 Olist sales with Webshoppers 38 market findings.

Highlights alignment and differences across product categories.

Uses both analytics and document evidence.

Demo 04 | Evidence boundaries

A question about an absent HR policy should not produce an invented policy.



OBSERVED BEHAVIOR

No annual-leave policy was found in the available documents.

NEXORA explicitly declined to invent an answer.

Tool used: search_enterprise_documents.

Engineering depth | Problems solved

Implementation choices and incidents encountered during development.

Resumable ingestion

Checkpoint recovery for interrupted multimodal indexing.

Figure enrichment

115 figure chunks selectively enriched in an earlier indexing run.

Retrieval debugging

Targeted experiment corrected a missed-chart retrieval case.

SQL service incident

Diagnosed Azure SQL 40613 transient unavailability; retry deployment unverified.

Technology stack | Azure-first engineering

Technologies shown in the project source and deployment materials.

EXPERIENCE

Next.js · React · TypeScript · MSAL

API + AGENTS

Python · FastAPI · GPT tool calling

DATA + SEARCH

Azure SQL · Azure AI Search · Document Intelligence

SECURITY

Microsoft Entra ID · managed identity · RBAC

DEPLOYMENT

Azure Container Apps · ACR · Docker · Bicep

QUALITY

pytest · GitHub Actions · targeted retrieval tests

NEXORA

From Azure architecture to a working AI application.

Demonstrated: SQL analytics · document RAG · combined reasoning · evidence boundaries

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github.com/zyani-chaimaa/azure-agentic-enterprise-copilot